Prayush B Menon

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OBJECTIVE

Embedded Systems & Firmware Engineer with an ECE background and hands-on expertise in bare-metal C development on PIC microcontrollers. Proficient in serial communication protocols (CAN, UART, SPI, I2C), Linux systems programming (POSIX, IPC, signals), memory-efficient data structures, and GDB debugging. Experienced in low-level firmware architecture, hardware-software integration, and software test automation.

TECHNICAL SKILLS

Languages: C / C++, Java
Microcontrollers & Protocols: PIC18F4580, CAN, UART, I2C, SPI
Systems & Concepts: POSIX System Calls, IPC (Pipes, Signals), Multiprocessing, Memory Management, Bit Manipulation
Developer Tools & OS: Linux (Fedora/Ubuntu), GDB, Make, Git, Neovim, Vim, Postman, Burp Suite
Software Testing & Management: Selenium, Playwright, Performance Monitoring (Lighthouse), Test Automation, JIRA

PROJECTS

Minishell (Unix Command Shell) GitHub

C, Linux Systems Programming, POSIX APIs, Signals, Process Management

- Built a Unix command shell in C supporting command execution, dynamic prompts, built-in commands (`cd`, `pwd`, `exit`), and multi-stage pipelines using `pipe()` and `dup2()`.
- Implemented signal handling for `SIGINT`, `SIGTSTP`, and `SIGCHLD` with a linked-list job table to manage background tasks and reap zombie processes via `waitpid()`.

Multithreaded TFTP Server & Client GitHub

C, UDP Sockets, POSIX Threads, Semaphores, Network Protocols

- Built a multithreaded TFTP file transfer server and client in C over UDP sockets, handling concurrent client uploads and downloads with POSIX worker threads.
- Limited active client transfers to 10 using a counting semaphore (`sem_trywait`) that sends instant busy replies, assigning each transfer to a dedicated ephemeral port with binary octet mode.

Concurrent HTTP/1.1 Server GitHub

C, POSIX Sockets, pthreads, Semaphores, HTTP/1.1, Systems Programming

- Architected a concurrent HTTP/1.1 server in C using an 8-worker thread pool and a thread-safe ring buffer synchronized with `pthread_mutex_t` and counting semaphores (`sem_t`).
- Implemented dynamic URL routing, header extraction, and static file delivery with automatic MIME detection, alongside disk-backed `POST` file uploads and zero socket leaks.

Car Black Box (Event Data Recorder) GitHub

Embedded C, PIC18F4580, I2C, UART, ADC, EEPROM, RTC, CLCD

- Wrote bare-metal I2C functions for a PIC18F4580 microcontroller to log gear shifts, clock timestamps, and ADC speed values to external EEPROM.
- Implemented a circular buffer to store and retrieve the 10 latest events, with a 16x2 CLCD menu, digital keypad authentication, and UART data export to PC.

Car Dashboard (Automotive CAN Simulator) GitHub

Embedded C, PIC18F4580, CAN Protocol, LCD, Embedded Systems

- Built a two-node automotive dashboard simulator using PIC18F4580 microcontrollers communicating over an isolated CAN bus.
- Transmitted real-time vehicle telemetry (speed, RPM, gear) from the engine ECU and decoded incoming CAN frames to update a live LCD dashboard using hardware filter masks.

Lexical Analyzer for C GitHub

C, Lexical Analysis, Tokenization, Binary Search, File I/O

- Built a C source code tokenizer that strips single/multi-line comments and preprocessor directives, classifying code into keywords, identifiers, constants, and operators.
- Used binary search over sorted keyword arrays for fast lookups, reporting line/column error locations for unclosed quotes and unmatched parentheses or braces.

Red-Black Tree (Self-Balancing BST) GitHub

C, Data Structures, Algorithms, Tree Rotations, Memory Management

- Engineered a self-balancing Red-Black Binary Search Tree in C based on CLRS algorithms, guaranteeing $O(\log N)$ worst-case time complexity for search, insertion, and deletion.
- Implemented left/right tree rotations, recoloring fixups for insertion/deletion, and explicit sentinel (`NIL`) nodes to eliminate null-pointer checks during balancing.

Arbitrary Precision Calculator (APC) GitHub

C, Data Structures, Algorithms, Dynamic Memory Allocation

- Built an arbitrary-precision calculator in C using doubly linked lists to perform arithmetic on arbitrarily large numbers without 64-bit integer limits.
- Implemented addition, subtraction, multiplication, and division with manual carry/borrow logic across nodes, digit compression, and zero memory leaks.

Image Steganography (LSB) GitHub

C, Bit Manipulation, Binary File I/O, Image Processing

- Built a command-line tool in C to hide and extract text messages inside 24-bit BMP images using the Least Significant Bit (LSB) technique.
- Embedded a 2-byte magic string and payload length while preserving the 54-byte BMP header to keep images visually indistinguishable and ensure safe decoding.

MP3 Tag Reader & Editor GitHub

C, Binary File Parsing, ID3v2 Tags, File I/O

- Built a command-line tool in C to view and edit ID3v2.3 metadata (title, artist, album) directly inside MP3 audio files.
- Parsed binary frame headers, handled big-endian byte offsets, and updated tag text in-place with frame size recalculation to preserve audio integrity.

EXPERIENCE

Senior Quality Engineer I July 2024 – January 2026

Fresh Prints Remote

- Built an automated frontend performance testing infrastructure with Lighthouse, reducing manual testing time by 80% through bi-weekly automated performance monitoring across environments.
- Led backend data migration validation for 7,000+ design records from Shopify to Directus CMS, developing custom validation scripts that ensured 100% data integrity and schema consistency.
- Identified critical P1 security vulnerabilities using Burp Suite to detect unauthorized access vectors, collaborating directly with development teams for immediate patch verification.

QA Automation Engineer June 2023 – July 2024

Cognizant Technology Solutions Bangalore, India

- Developed and maintained test automation frameworks using Java and Selenium WebDriver, standardizing regression testing across core application modules.
- Triaged and tracked software defects across agile sprint cycles, performing root-cause analysis to verify fixes and ensure release reliability.
- Reviewed technical requirements and system specifications to author automated test plans, expanding test coverage across end-to-end user workflows.
- Collaborated with engineering teams to optimize test execution pipelines, migrating manual validation suites into continuous automated test runs.

QA Engineer June 2022 – June 2023

Cognizant Technology Solutions Bangalore, India

- Executed functional and regression test suites across multi-tier applications, isolating defect root causes and reducing turnaround time for issue resolution.
- Performed REST API validation and endpoint testing using Postman, verifying payload schemas, HTTP status codes, and backend data flows.
- Authored and maintained comprehensive test suites covering product functional specifications, mitigating critical defect escapes into production releases.
- Scripted test automation routines to replace repetitive manual verification, accelerating regression testing cycles while maintaining consistent test repeatability.

EDUCATION & TRAINING

Emertxe Information Technologies Bangalore, India

Advanced Embedded Systems Training (PIC18F4580, C/C++, Protocols, Linux) 2025 – 2026

Amrita School of Engineering Kollam, Kerala

B.Tech in Electronics and Communications Engineering (CGPA: 8.1/10) June 2018 – August 2022